Exponential sample lower bounds for sparse parity under a globally separated neural prior

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Abstract

We study a fixed-depth Bayesian neural network with Gaussian weights, hyperbolic-tangent hidden activations, and a linear output, conditioned to have a fixed absolute output margin on every Boolean input and then on noiseless training labels. For every fixed width-scaling exponent $0 < \gamma < 1$, we prove an $\exp(\Omega(k))$ sample lower bound for learning parity on any k coordinates of a d-dimensional input, uniformly for growing $k \le d$. With probability at least $3/4$ over the training sample, every sign function selected in the width-first limit has test error at least $3/8$. Consequently, at least $1/16$ of inputs lack correct-label posterior probability greater than $2/3$, regardless of the actual mixture weights among tied minimizers. A random-restriction argument shows that a low-degree polynomial approximating a globally separated accurate parity predictor must have exponentially large squared amplitude in the ratio of parity degree to polynomial degree. We combine this with the paired Fourier spectrum of the limiting kernel and uniform bounds on sampled tail covariance matrices. A second argument improves the sample range to $\exp\{c \max[k, \sqrt{k}\log(e(d+1)/\sqrt{k})]\}$ in very large ambient dimensions. All width-limit, spectral, and interpolation arguments are included; the one-logarithm sampling estimate uses an intermediate inequality from the proof of a lemma of Lee, and a random principal-submatrix bound of Tropp, with the needed modifications indicated. A generalization upper bound has scale $\exp\{O(k\log(ed/k))\}$ for growing sparse k. This rules out a literal $d^{\Omega(k)}$ lower bound uniformly over all $k = o(d)$; a lower bound on the natural $\exp\{\Omega(k\log(ed/k))\}$ scale remains open.

1 Model and main result

Let $\mathcal{X} = \{-1, 1\}^d$ with uniform probability measure. Fix $U \subseteq [d]$, $|U| = k$, and let $f(x) = \chi_U(x) = \prod_{j\in U} x_j$. The learner is not given U. All hidden layers have width N, the number of weight matrices $L \ge 2$ is fixed, and

$$h^0(x) = (1, x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \le s < L), \qquad z_N(x) = W^L h^{L-1}(x).$$

There are no further biases. All weights are independent centered Gaussians with variances

$$\operatorname{Var} W^1_{ij} = N^{1-\gamma}/d, \qquad \operatorname{Var} W^s_{ij} = N^{-\gamma} \quad (1 < s < L), \qquad \operatorname{Var} W^L_i = N^{-\gamma-1}. \tag{1}$$

Fix $a > 0$, for example $a = 1/2$, and a fixed $0 < \gamma < 1$. Condition the prior on $|z_N(x)| > a$ for every input x. For independent uniform training inputs $S = (X_1, \dots, X_n)$, condition further on $f(X_i)z_N(X_i) > 0$ for every i. Define the actual correct-label probability and its confidence failure by

$$q_{N,S}(x) = \mathbb{P}\{f(x)z_N(x) > 0 \mid |z_N(y)| > a \ \forall y;\ f(X_i)z_N(X_i) > 0 \ \forall i\},$$
$$F_N(S) = \mathbb{P}_x\{q_{N,S}(x) \le 2/3\}.$$

A successful prediction therefore requires strictly more than 2/3 probability of the correct label. The test input is independent and uniform.

Put $D = d + 1$, $\beta = 2^{-(L-1)}$, and

$$B_{d,k} = \sqrt{k} \log \frac{eD}{\sqrt{k}}, \qquad W_{d,k} = \max\{k, B_{d,k}\}. \tag{2}$$

Theorem 1.1 (Actual-posterior sample lower bound)**.** *For each fixed $L \geq 2$ and $a > 0$, there are $c > 0$, k_0, and d_0 such that, for every $d \geq d_0$, $k_0 \leq k \leq d$, support U of size k, integer*

$$0 \leq n \leq \exp(cW_{d,k}),$$

and fixed $0 < \gamma < 1$,

$$\mathbb{P}_S \left\{ \liminf_{N \to \infty} F_N(S) \geq \frac{1}{16} \right\} \geq \frac{3}{4}.$$

The conditioning events have positive probability for all sufficiently large widths at every fixed dimension and training sample.

The width limit is taken first, with d, k, L, a, γ, S fixed. No simultaneous width–dimension limit or quantified minimum width is asserted. The training-sample probability holds for each specified n, with constants uniform over the displayed range; no monotonicity of fixed-target posterior risk is used.

The lower bound is exponential in the number k of relevant coordinates even when $k/d \to 0$. The extra $B_{d,k}$ term is useful when the ambient dimension is very large relative to k. For k/d bounded below, the theorem gives an $\exp(\Omega(d))$ sample lower bound. When $d/k \to \infty$, it does not reach $\exp\{ck \log(ed/k)\}$. In particular it does not resolve a $d^{\Omega(k)}$ lower bound in a polynomially sparse regime.

1.1 Proof structure and notation

Write $\Psi(t) = 2 \arcsin(t)/\pi$ and

$$K(x, y) = \Psi^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d + 1} \right).$$

The actual network has a Gaussian large-deviation rate with this covariance. Conditioning on the global margin and the labels therefore selects compatible sign functions of minimum energy $\|z\|_K^2/2 = z^{\mathsf{T}} K^{-1} z/2$. We prove a simultaneous error lower bound for every minimizer, so the argument applies to arbitrary actual weights among tied minimizers.

All function L^p norms and Walsh coefficients use the uniform cube probability measure. The characters diagonalize the normalized covariance operator $2^{-d}K$; let λ_j denote its degree-j eigenvalue. Thus

$$\|z\|_K^2 = \sum_{V \subseteq [d]} \frac{\widehat{z}(V)^2}{\lambda_{|V|}}.$$

Put $D_r = \sum_{j \leq r} \binom{d}{j}$, let $Q_r = K_{>r}$ retain the Fourier degrees above r, and write $\tau_r = Q_r(x, x) = \sum_{j > r} \binom{d}{j} \lambda_j$. For an arbitrary kernel J, J_{SS} is its sample Gram matrix.

Localized interpolation bounds every minimizing energy by $C(d + nd^{\beta})$. At the largest odd cutoff with $n \geq AD_r \log(D_r + 1)$, a one-sided squared sampling theorem controls the low-degree part of a minimizing field. The spectrum has equal eigenvalues at consecutive even and odd

levels. A near-orthogonal estimate controls the sampled tail covariance matrix at the last sampled cutoff. This bounds the low-degree amplitude by a polynomial in the ratio of ambient dimension to cutoff. The energy budget and the paired spectrum then make the part above the next paired level small in L^2 deterministically. A random restriction of the active coordinates turns this head bound into a test-error lower bound: restricting to about $(k/r)^2$ coordinates leaves almost all of a degree-r polynomial below the threshold degree needed to approximate parity, unless its amplitude is exponentially large in k/r. A direct polynomial-threshold argument supplies the additional $B_{d,k}$ range and handles the smallest samples.

2 The kernel rate of the actual network

Let $M = 2^d$, and identify output functions with vectors in $\mathbb{R}^M$. Define

$$\Psi(t) = \frac{2}{\pi}\arcsin t, \qquad K(x,y) = \Psi^{\circ(L-1)}\left(\frac{1+x\cdot y}{d+1}\right). \tag{3}$$

The positive definiteness of K is proved in the next section. For each Boolean function g, put

$$E_K(g) = \min_{g(x)z(x)\geq a\ \forall x} \frac{1}{2} z^T K^{-1} z. \tag{4}$$

The minimum exists since its feasible set is closed and nonempty and the objective is coercive. Replacing the closed inequalities by strict ones gives the same infimum, by scaling a feasible vector by $1+\epsilon$.

Proposition 2.1 (Posterior concentration on minimum-energy signs)**.** *For the actual network* (1)*, at every fixed* $0 < \gamma < 1$*,*

$$\lim_{N\to\infty} N^{-\gamma} \log \mathbb{P}\{g(x)z_N(x) > a\ \forall x\} = -E_K(g). \tag{5}$$

Consequently, after any fixed noiseless training sample, posterior mass tends to one on the compatible functions minimizing E_K*. No assertion about the relative weights of tied minimizers is needed.*

Proof. For hidden layer s, let $Q_N^s(x,y) = N^{-1}\sum_i h_i^s(x)h_i^s(y)$. We first establish, for each fixed $\epsilon > 0$,

$$\mathbb{P}\{\|Q_N^{L-1} - K\| > \epsilon\} \leq C_\epsilon e^{-c_\epsilon N} \tag{6}$$

for all sufficiently large N. Any matrix norm can be used, since M is fixed. Conditional on the preceding layer, the rows of the next hidden layer are independent bounded vectors. Hoeffding's inequality controls all their empirical second moments. Their preactivations have covariance $N^{1-\gamma}Q_N^{s-1}$; at the first layer the covariance is $N^{1-\gamma}(1+x\cdot y)/d$. Because $N^{1-\gamma}\to\infty$, the conditional second-moment map converges to the Gaussian-sign map

$$Q \longmapsto \left[\Psi\left(\frac{Q(x,y)}{\sqrt{Q(x,x)Q(y,y)}}\right)\right]_{x,y}.$$

This follows by dominated convergence for a Gaussian vector, since each scalar variance stays bounded away from zero in a neighborhood of the limiting covariance. The convergence is uniform in such a neighborhood: a compactness subsequence and a continuous square root coupling reduce the assertion to the same dominated-convergence argument. The limiting map is continuous there.

Inducting through the fixed number of layers, with smaller fixed tolerances at earlier layers, proves (6). The first-layer limit normalizes the diagonal $(d+1)/d$, giving the kernel in (3).

Conditionally on the hidden layers, the readout is exactly Gaussian:

$$Z_N \mid Q_N^{L-1} \sim \mathcal{N}(0, N^{-\gamma} Q_N^{L-1}). \tag{7}$$

Couple it to $Y_N = N^{-\gamma/2} K^{1/2} \xi$, where $\xi \sim \mathcal{N}(0, I_M)$ is independent of the hidden layers. On a fixed sufficiently small neighborhood of the positive-definite matrix K, the square-root map is Lipschitz. Hence, for $\delta > 0$,

$$\mathbb{P}\{\|Z_N - Y_N\| > \delta\} \le C_\epsilon e^{-c_\epsilon N} + \mathbb{P}\{\|\xi\| > \delta N^{\gamma/2}/(C\epsilon)\}.$$

First divide logarithms by N^γ and let $N \to \infty$; then let $\epsilon \downarrow 0$. The resulting upper limit is $-\infty$, since $\gamma < 1$. The coupled vectors are therefore exponentially equivalent at speed N^γ.

The fixed-dimensional Gaussian vectors Y_N have large-deviation rate $I(z) = z^T K^{-1} z/2$, directly from their densities. Exponential equivalence transfers the same bounds to Z_N. Applying the upper bound to a closed margin orthant and the lower bound to its interior gives (5), since their infima agree. Finally, there are only finitely many Boolean functions for fixed d. Dividing each compatible orthant probability by their sum proves the posterior concentration claim. □

This is a rate statement for the network itself. It does not require relative-error approximation of rare probabilities, nor convergence in total variation to a Gaussian posterior.

3 Fourier level masses of the kernel

Put $D = d+1$, $\beta = 2^{-(L-1)}$, and $F = \Psi^{\circ(L-1)}$. The characters $\chi_U(x) = \prod_{j\in U} x_j$ diagonalize the normalized covariance operator $M^{-1}K$, $M = 2^d$. Let λ_j be its eigenvalue at degree j. We first work on the full D-cube with kernel $F(u \cdot v/D)$, whose degree-j eigenvalue is denoted by a_j. Constants in this and the next section may depend on L. Notation $b \asymp c$ means two inequalities by positive constant multiples, uniformly in the stated ranges.

Lemma 3.1 (Power coefficients and paired eigenvalues)**.** *Write $F(z) = \sum_{m \text{ odd}} c_m z^m$. Every odd coefficient is positive, $\sum_m c_m = 1$, and*

$$c_m \asymp \frac{\Gamma(m-\beta)}{m!} \asymp m^{-1-\beta} \qquad (m \ge 1 \text{ odd}).$$

The full-cube eigenvalues vanish at even degrees and strictly decrease through odd degrees. They are positive at every odd degree. On the original d-cube,

$$\lambda_{2k} = \lambda_{2k+1} = a_{2k+1} > 0,$$

with a final unpaired eigenvalue if d is even. In particular K is positive definite.

Proof. The series of Ψ has positive coefficients at every odd exponent, with

$$[z^{2k+1}]\Psi(z) = \frac{2}{\pi} \frac{\binom{2k}{k}}{4^k(2k+1)}.$$

Composition preserves strict positivity and oddness. Near $z = 1$ in a slit neighborhood,

$$\Psi(z) = 1 - b(1-z)^{1/2} + O((1-z)^{3/2}), \qquad b = \frac{2\sqrt{2}}{\pi}.$$

Induction gives $F(z) = 1 - A(1-z)^\beta + O((1-z)^{3\beta})$, with $A > 0$; at each composition its leading constant changes from A_t to $b\sqrt{A_t}$, starting from $A_1 = b$. These expansions admit algebraic coefficient transfer [2, Chapter VI]. Here are the analytic-domain details. On the unit circle outside $\{-1, 1\}$, positivity of every odd coefficient gives $|\Psi(z)| < 1$: equality in the triangle inequality would require $z^2 = 1$. Hence the iterated functions extend analytically through every compact arc away from these endpoints. At 1, restrict $\arg(1-z)$ to a closed proper subinterval of $(-\pi, \pi)$. Each square-root composition reduces this angular range, and its positive leading coefficient keeps its image in the slit neighborhood for the next arcsine composition. Oddness gives the identical argument at -1. These local sectors and the compact arcs provide the required two-singularity indented domain. The contributions of 1 and -1 add at odd exponents, giving

$$c_m \sim \frac{2A}{-\Gamma(-\beta)} m^{-1-\beta} \quad (m \to \infty \text{ odd}).$$

Together with positivity of the finitely many initial coefficients and the gamma-ratio asymptotic, this proves the coefficient comparisons. For $L = 2$ the explicit formula and Stirling's bounds suffice instead.

Let $p_m(j)$ be the probability that the simple coordinate-flip walk on the D-cube reaches a specified distance-j vertex in m steps. Expanding $(u \cdot v/D)^m$ shows that its degree-j Fourier coefficient is $p_m(j)$. Thus $a_j = \sum_{m \text{ odd}} c_m p_m(j)$. For the walk recursion,

$$\begin{aligned} D[p_{m+1}(j) - p_{m+1}(j+2)] = j[p_m(j-1) - p_m(j+1)] \\ + (D-j-2)[p_m(j+1) - p_m(j+3)]. \end{aligned}$$

Induction from time zero gives nonincreasing probabilities through each parity class; terms with zero boundary coefficients are omitted. The term $m = j$ makes the odd-eigenvalue inequalities strict and proves positivity. Even-degree eigenvalues vanish by parity. Restricting the distinguished coordinate to $+1$ gives $\lambda_q = a_q + a_{q+1}$, proving the claims. □

For odd r, the exact combined multiplicity of reduced degrees $j-1, j$ is $\binom{d}{j-1} + \binom{d}{j} = \binom{D}{j}$. Consequently

$$D_r = \sum_{\substack{j \le r \\ j \text{ odd}}} \binom{D}{j}, \quad \tau_r = \sum_{\substack{j > r \\ j \text{ odd}}} \binom{D}{j} a_j, \quad \lambda_{r+1} = a_{r+2}. \tag{8}$$

The odd cutoff is essential for these exact identities.

Lemma 3.2 (Sharp level and tail estimates)**.** *Uniformly for odd $1 \le j \le D/4$ and odd $1 \le r \le D/8$,*

$$\binom{D}{j} a_j \asymp j^{-1-\beta}, \tag{9}$$

$$\tau_r \asymp r^{-\beta}, \tag{10}$$

$$\frac{\lambda_{r+1}}{\tau_r} \asymp \frac{1}{r\binom{D}{r+2}}, \tag{11}$$

$$s_{2,r} := \sum_{q > r} \binom{d}{q} \lambda_q^2 \le \frac{C r^{-2\beta-2}}{\binom{D}{r+2}}. \tag{12}$$

The same level-mass estimate holds up to $j \le \rho D$ for any fixed $\rho < 1/2$, with constants depending on ρ.

Proof. The gamma integral and Lemma 3.1 give, for odd j,

$$a_j \asymp D^{-\beta} \int_0^\infty u^{-1-\beta} p(u)^j (1-p(u))^{D-j}\, du, \quad p(u) = \frac{1-e^{-2u}}{2}, \tag{13}$$

$$\asymp D^{-\beta} \int_0^{1/2} p^j (1-p)^{D-j} w_\beta(p)\, dp, \tag{14}$$

where $w_\beta(p) = [-\frac{1}{2}\log(1-2p)]^{-1-\beta}/(1-2p)$. Indeed the gamma integral Poissonizes the walk time, and independent Poisson coordinate counts give independent coordinate parities with success probability $p(u)$. Since an odd-distance vertex has zero probability at even walk times, no extra parity factor occurs. The weight w_β is integrable near $1/2$ and comparable to $p^{-1-\beta}$ on $[0, b]$ for every fixed $b < 1/2$.

Fix $\rho < b < 1/2$ and take $j \le \rho D$. The integral on $[0, b]$ is comparable to the complete beta integral $B(j-\beta, D-j+1)$. For the lower comparison, a beta random variable with these parameters has mean at most ρ, so Markov's inequality puts at least a fraction $1-\rho/b$ of its mass in $[0, b]$. The identity

$$\binom{D}{j} B(j-\beta, D-j+1) = \frac{\Gamma(D+1)\Gamma(j-\beta)}{\Gamma(j+1)\Gamma(D+1-\beta)} \asymp D^\beta j^{-1-\beta}$$

gives the desired mass. On $[b, 1/2]$, the binomial factor decreases with p and, after multiplication by $\binom{D}{j}$, is at most $\exp[-D\,\mathrm{KL}(\rho\|b)]$. The remaining integrable weight makes this contribution exponentially small. This proves (9) and its stated extended range.

Sum (13) over the odd distances above r. Dropping the parity restriction and using the binomial Markov bound gives

$$\tau_r \le C D^{-\beta} \int_0^\infty u^{-1-\beta} \min\{1, Du/(r+1)\}\, du \le C(r+1)^{-\beta}.$$

For the lower bound, sum (9) over the odd j in $(r, 2r+2]$, using a fixed cutoff strictly below $D/2$ as needed at the endpoint. This yields $cr^{-\beta}$; bounded r are absorbed by a change of constants. Equation (11) follows from (9), (10), and (8).

For the second moment, use paired multiplicities to write $s_{2,r} = \sum_{j>r,\ j \text{ odd}} \binom{D}{j} a_j^2$. Up to an odd cutoff b_D near $D/4$, its terms are at most $Cj^{-2\beta-2}/\binom{D}{j}$. Consecutive odd inverse binomial coefficients have ratio

$$\frac{(j+1)(j+2)}{(D-j)(D-j-1)},$$

uniformly less than a fixed constant below one in this range. The sum is therefore bounded by a constant times its first term, at $j = r+2$. Beyond b_D, monotonicity gives the bound $a_{b_D}\tau_{b_D} \le CD^{-2\beta-1}/\binom{D}{b_D}$, which is exponentially smaller than the claimed bound uniformly for $r \le D/8$. This proves (12). □

Finally, two-step binomial ratios give, uniformly for odd $r \le D/8$,

$$\binom{D}{r} \le D_r \le C\binom{D}{r}, \qquad \frac{D_r}{\binom{D}{r+2}} \le C(r/D)^2. \tag{15}$$

All constants used here can be fixed independently of a subsequently chosen small positive limit $r/d \to \alpha$.

Corollary 3.3 (Lower bounds used by the interpolation)**.** *For sufficiently large d,*

$$\lambda_0 \ge c/d, \qquad \lambda_j \ge \frac{cd^{-1-\beta}}{\binom{d}{j}} \quad (0 \le j \le .49d), \qquad v := \lambda_{\min}(K) \ge c(d\log d)^{-\beta}.$$

Also $v \to 0$.

Proof. The first bound follows from the level estimate at full-cube degree one. For odd reduced degree j, use $\lambda_j = a_j$ and $\binom{D}{j}/\binom{d}{j} = D/(D-j)$. For even j, use $\lambda_j = a_{j+1}$ and $\binom{D}{j+1}/\binom{d}{j} = D/(j+1)$. The extended level bound with any fixed cutoff between .49 and 1/2 then gives the stated uniform lower bound, including $j = 0$.

For any odd full-cube degree $j \le D$, restrict the integral (13) to $u \ge \frac{1}{2}\log(2D)$. In this range $p(u) \ge (1 - 1/(2D))/2$ and $1 - p(u) \ge 1/2$, so $p(u)^j(1-p(u))^{D-j} \ge c2^{-D}$. Integrating $u^{-1-\beta}$ gives $a_j \ge c2^{-D}D^{-\beta}(\log D)^{-\beta}$. Every reduced eigenvalue equals an odd a_j, and the matrix eigenvalues are M times the normalized ones. Thus $v \ge c(d\log d)^{-\beta}$. Finally, for any fixed small positive cutoff fraction, the minimum normalized eigenvalue is at most the average in its tail, whence $v \le M\tau_r/(M - D_r) = O(d^{-\beta})$. This proves $v \to 0$. □

4 A globally separated interpolation comparator

The next construction bounds the energy of every minimizer. It works for arbitrary training labels and enforces the margin on the entire cube.

Lemma 4.1 (Localized interpolation)**.** *There is d_L such that, for $d \ge d_L$ and $1 \le n \le e^{d/25}$, with probability at least $1 - e^{-c_L d}$ over independent uniform inputs, the following holds simultaneously for every labeling $y_i \in \{-1, 1\}$ and every $A > 0$. There is a field h with*

$$|h(x)| > A \quad \text{for every } x, \qquad y_i h(X_i) > A \quad \text{for every } i, \qquad \|h\|_K^2 \le C_L A^2(d + nd^\beta).$$

For $n = 0$, the same deterministic energy bound holds using a constant field.

Proof. For $z \in \mathcal{X}$ define the nonnegative bump

$$b_z(x) = 20\int_{2/5}^{9/20} (1-2p)^{\mathrm{dist}(x,z)}\, dp.$$

It satisfies $b_z(z) = 1$ and $b_z(x) \le (1/5)^{\mathrm{dist}(x,z)}$. Its degree-$j$ Walsh coefficients, apart from their translation signs, are

$$\widehat{b}_j = 20\int_{2/5}^{9/20} (1-p)^{d-j}p^j\, dp, \qquad \binom{d}{j}\widehat{b}_j \le \frac{20}{d+1}, \qquad \widehat{b}_j \le (3/5)^d.$$

The middle inequality follows by extending to the complete beta integral. By Corollary 3.3, $\lambda_j \ge c_L d^{-1-\beta}/\binom{d}{j}$ for $j \le .49d$. Hence these degrees contribute at most

$$C_L d^{1+\beta} \sum_{j \le .49d} \left[\binom{d}{j}\widehat{b}_j\right]^2 \le C_L d^\beta$$

to $\|b_z\|_K^2$. For $j > .49d$, the integrand is increasing in p on the integration interval, so $\widehat{b}_j \le .55^{d-j}.45^j$. Let $v = \lambda_{\min}(K)$ be the smallest matrix eigenvalue. The same corollary gives $v \ge c_L(d\log d)^{-\beta}$ and $\lambda_j \ge v/2^d$. Thus the remaining energy is at most

$$\frac{2^d}{v}\sum_{j>.49d}\binom{d}{j}.55^{2(d-j)}.45^{2j} \le \mathrm{poly}(d)e^{-c_* d},$$

where

$$c_* = -[\log 2 + \mathsf{h}(.49) + 1.02\log(.55) + .98\log(.45)] > .0062, \qquad \mathsf{h}(t) = -t\log t - (1-t)\log(1-t).$$

The exponent decreases beyond $.49d$, so the binomial entropy bound applies. We have proved $A_d := \|b_z\|_K^2 \le C_L d^\beta$.

Put $J(z,w) = \langle b_z, b_w\rangle_K$. Its normalized Fourier eigenvalues are $\nu_j = \widehat{b}_j^2/\lambda_j \le .72^d/v$. Parseval and Cauchy–Schwarz give

$$\mathbb{E}_{X,Y}|J(X,Y)| \le \left(\sum_j \binom{d}{j}\nu_j^2\right)^{1/2} \le \sqrt{\nu_{\max}A_d}.$$

For every subset T of the sampled indices, including one chosen after the sample is seen,

$$\left\|\sum_{i\in T} b_{X_i}\right\|_K^2 \le nA_d + \sum_{i\ne j}|J(X_i,X_j)|.$$

The cross sum has expectation at most $n(n-1)\sqrt{\nu_{\max}A_d}$. Markov's inequality at threshold nd^β has exponentially small failure probability for $n \le e^{.04d}$, because $\frac{1}{2}\log(1/.72) > .16$. Thus all subsets simultaneously have energy at most $C_L n d^\beta$.

Also, except with exponentially small probability, every pair of different sample indices has inputs at Hamming distance greater than $.2d$. A union bound costs at most

$$n^2\exp[-d(\log 2 - \mathsf{h}(.2))], \qquad \log 2 - \mathsf{h}(.2) > .19.$$

This event excludes repeated inputs. At any input at most one center is within distance $.1d$; the sum of all other bumps is at most $n(.2)^{.1d} = o(1)$. Consequently, for every selected subset of centers, its bump sum is $1 + o(1)$ at a selected center and at most $.2 + o(1)$ everywhere else.

For $T = \{i : y_i = -1\}$, take

$$h = 2A\left(1 - 2\sum_{i\in T} b_{X_i}\right).$$

The preceding estimates prove its strict global and training margins for all sufficiently large d. Finally $\|1\|_K^2 = 1/\lambda_0 \le C_L d$ by Corollary 3.3; the squared triangle inequality proves the asserted energy bound. For $n = 0$, use $h = 2A$. □

5 One-sided squared sampling with one logarithm

The sampling estimate needed below concerns a squared one-sided loss. It follows from a metric estimate in the proof of Lee's chaining theorem [3, Lemma 2.12, equation (2.20)]. We spell out the comparison: a first-power ReLU sampling theorem alone would not give this result. Throughout this section, C denotes a universal constant that may change from line to line, and $t_- := \max\{-t, 0\}$.

Lemma 5.1 (A squared Lipschitz process)**.** *Let $a_1, \dots, a_m \in \mathbb{R}^D$, and set*

$$R = \max_{i \le m} \|a_i\|_2, \qquad V = \sup_{\|u\|_2 \le 1} \left(\sum_{i=1}^m \langle a_i, u \rangle^2 \right)^{1/2}.$$

For each i, let $\psi_i : \mathbb{R} \to \mathbb{R}$ satisfy $\psi_i(0) = 0$ and $\mathrm{Lip}(\psi_i) \le 1$. For independent uniform signs ϵ_i,

$$\mathbb{E}_\epsilon \sup_{\|u\|_2 \le 1} \left| \sum_{i=1}^m \epsilon_i \psi_i(\langle a_i, u \rangle)^2 \right| \le CR\sqrt{\log(2(m+D))}\, V. \tag{16}$$

Proof. Use the genuine seminorms $N_i(u) = |\langle a_i, u \rangle|$ and the symmetric quasimetric

$$d_A(u,v)^2 = \sum_{i=1}^m N_i(u-v)^2 (N_i(u)^2 + N_i(v)^2).$$

The stronger intermediate estimate established in the proof of [3, Lemma 2.12] is

$$\gamma_2(T, d_A) \le C(\lambda\sqrt{\log(2D)} + \kappa) \sup_{u \in T} \left(\sum_i N_i(u)^2 \right)^{1/2}, \qquad T \subseteq B_2^D, \tag{17}$$

where

$$\lambda = \max_i \left(\mathbb{E} N_i(g)^2\right)^{1/2}, \qquad \kappa = \mathbb{E} \max_i N_i(g), \qquad g \sim \mathcal{N}(0, I_D).$$

Here γ_2 is the generic-chaining functional. This is equation (2.20) of that proof, with the energy normalization (2.17) undone. Seminorms follow from the norm statement by replacing $N_i(u)$ with $(N_i(u)^2 + \varepsilon^2 \|u\|_2^2)^{1/2}$ and then letting $\varepsilon \downarrow 0$; the resulting metrics dominate d_A, and the parameters on the right converge. For our rank-one seminorms, $\lambda = R$ and $\kappa \le R\sqrt{2\log(2m)}$.

Let $X_u = \sum_i \epsilon_i \psi_i(\langle a_i, u \rangle)^2$. Its subgaussian increment metric is bounded by $\sqrt{2}\, d_A$, because

$$\begin{aligned} &\sum_i \left(\psi_i(\langle a_i, u \rangle)^2 - \psi_i(\langle a_i, v \rangle)^2 \right)^2 \\ &\quad \le \sum_i |\langle a_i, u - v \rangle|^2 \left(|\langle a_i, u \rangle| + |\langle a_i, v \rangle| \right)^2 \le 2 d_A(u,v)^2. \end{aligned}$$

We used both Lipschitz continuity and $|\psi_i(t)| \le |t|$. The subgaussian chaining inequality, together with $X_0 = 0$, bounds $\mathbb{E} \sup X_u$ by $C\gamma_2(B_2^D, d_A)$. Apply the same argument to $-X_u$ and add. Equation (17) now proves (16). □

Remark 5.2 (The normalization in the metric estimate)*.* The proof of [3, Lemma 2.12] passes through a coarse additive constant in its equation (2.26), after normalizing $\sup_T \sum_i N_i^2 = 1$. The homogeneous version (17) follows by retaining the scale in the two geometric terms of that argument. Indeed, in the rank-one case $T \subseteq B_2^D$ implies $N_i(u) \le \lambda$ on T; writing $\|u\|_{\mathcal{N}} = \max_i N_i(u)$ gives $d_A(u,v) \le \sqrt{2}\|u - v\|_{\mathcal{N}}$ on T. Thus $\mathrm{diam}_{d_A}(T) \le C\lambda$ and the volumetric packing constant in its Corollary 2.11 can be taken to be $C\lambda$, since $T \subseteq \lambda B_{\mathcal{N}}$. The additive term is therefore $C\lambda$, absorbed by $C(\lambda\sqrt{\log(2D)} + \kappa)$. The growth-functional portion of the proof is unchanged. If the packing constant is less than one, first rescale the quasimetric to apply that corollary in its stated normalization, and then scale back. Most importantly, throughout that argument the N_i are symmetric seminorms. No assertion that a positive part is a seminorm is used.

Lemma 5.3 (Uniform one-sided squared sampling)**.** *Let (Z, Y) be a random pair with $Z \in \mathbb{R}^D$, $Y \in \{-1, 1\}$, $\mathbb{E}ZZ^\mathsf{T} = I_D$, and $\|Z\|_2^2 = D$ almost surely. For independent copies (Z_i, Y_i), let*

$$\Delta_n = \sup_{\|u\|_2 \le 1} \left| \frac{1}{n} \sum_{i=1}^{n} (Y_i \langle Z_i, u \rangle)_-^2 - \mathbb{E}(Y \langle Z, u \rangle)_-^2 \right|, \qquad h_n = \sqrt{\frac{D \log(2(n+D))}{n}}.$$

Then $\mathbb{E}\Delta_n \le C h_n (1 + C h_n)$. In particular, for any fixed $\varepsilon, \delta > 0$ there is a finite $A_{\varepsilon,\delta}$ such that

$$n \ge A_{\varepsilon,\delta} D \log(2D) \quad \Longrightarrow \quad \mathbb{P}\{\Delta_n > \varepsilon\} \le \delta.$$

The labels may depend arbitrarily on the features.

Proof. Put $Q_n = n^{-1} \sum_i Z_i Z_i^\mathsf{T}$. Conditional on the observations, apply Lemma 5.1 to $a_i = Y_i Z_i / \sqrt{n}$ and $\psi_i(t) = t_-$. Here $R = \sqrt{D/n}$ and $V = \sqrt{\|Q_n\|_{\mathrm{op}}}$. Symmetrization gives

$$\mathbb{E}\Delta_n \le C h_n \, \mathbb{E}\sqrt{\|Q_n\|_{\mathrm{op}}}. \tag{18}$$

We also apply the same process lemma with $\psi_i(t) = t$. Since $\mathbb{E}Q_n = I_D$, symmetrization and Jensen's inequality imply, with $A = \mathbb{E}\|Q_n\|_{\mathrm{op}}$,

$$A \le 1 + \mathbb{E}\|Q_n - I_D\|_{\mathrm{op}} \le 1 + C h_n \mathbb{E}\sqrt{\|Q_n\|_{\mathrm{op}}} \le 1 + C h_n \sqrt{A}.$$

Solving this quadratic inequality yields $\sqrt{A} \le 1 + C h_n$, after increasing C. Substitute into (18). Finally, at $n \ge A_{\varepsilon,\delta} D \log(2D)$ the quantity h_n can be made uniformly small by increasing $A_{\varepsilon,\delta}$. Markov's inequality proves the probability statement. It is a statement for each specified sample size n. □

6 A polynomial-threshold obstruction

The following elementary polynomial-threshold obstruction is due to Aspnes, Beigel, Furst, and Rudich [1]; we include its proof.

Lemma 6.1 (Polynomial thresholds cannot approximate parity)**.** *On a cube of dimension k, for $0 \le s < k$, every degree-s polynomial threshold, with a fixed common convention for zero ties, has full-parity error at least*

$$2^{-k} \sum_{j=0}^{\lfloor (k-s-1)/2 \rfloor} \binom{k}{j}. \tag{19}$$

This lower bound tends to $1/2$ whenever $s = o(\sqrt{k})$.

Proof. A sufficiently small constant perturbation resolves any zero ties without changing the specified threshold classifier. Thus suppose P has no zeros. Put $t = \lfloor (k - s - 1)/2 \rfloor$. If its error set had fewer than $\sum_{j \le t} \binom{k}{j}$ points, a nonzero polynomial Q of degree at most t could be chosen to vanish on every error. The function fPQ^2 would then be nonnegative everywhere and positive somewhere. But PQ^2, after multilinear reduction, has degree at most $s + 2t < k$, so its parity Fourier coefficient $\mathbb{E}fPQ^2$ is zero, a contradiction. This proves (19). The last assertion follows from the central binomial bound $\max_j 2^{-k} \binom{k}{j} = O(k^{-1/2})$. □

7 A degree–amplitude inequality from random restrictions

The next lemma uses the global margin to strengthen a degree-only obstruction. Its proof combines random restrictions with the parity approximation bound of Aspnes, Beigel, Furst, and Rudich.

Lemma 7.1 (Robust degree–amplitude inequality)**.** *Let $f = \chi_U$ on the uniform d-cube, where $|U| = k$, and let $|z(x)| \ge a > 0$ everywhere. Suppose $z = P + T$, where $\deg P \le u$, $1 \le u \le k$, and P, T have disjoint Fourier supports. There are absolute constants $c, C > 0$ such that*

$$\Pr\{\operatorname{sign} z \neq f\} \ge \frac{7}{16} - \frac{\|T\|_2^2}{a^2} - C\frac{\|P\|_2^2}{a^2}\exp\!\left(-c\frac{k}{u}\right). \tag{20}$$

More precisely, for integers $1 \le m \le k$ and $0 \le q < m$ the right-hand side can be replaced by

$$2^{-m}\sum_{j=0}^{\lfloor (m-q-1)/2\rfloor}\binom{m}{j} - \frac{\|T\|_2^2}{a^2} - \frac{\|P\|_2^2}{a^2}\min\left\{1, \left(\frac{eum}{k(q+1)}\right)^{q+1}\right\}. \tag{21}$$

Proof. Choose a uniform m-element subset $J \subseteq U$, fix the coordinates outside J uniformly, and truncate the restricted polynomial P to degree q in the remaining coordinates. Denote the resulting polynomial by Q. Viewed on the original cube, this operation retains exactly the Fourier terms $\widehat{P}(S)\chi_S$ with $|S \cap J| \le q$. Orthogonality therefore gives

$$\mathbb{E}_J\|z - Q\|_2^2 = \|T\|_2^2 + \sum_S \widehat{P}(S)^2 \Pr_J\{|S \cap J| > q\}.$$

Since $|S| \le u$, the probability is bounded by the upper tail of a hypergeometric random variable with population size k, at most u marked elements, and sample size m. The factorial-moment estimate gives

$$\Pr\{|S \cap J| \ge q+1\} \le \min\left\{1, \left(\frac{eum}{k(q+1)}\right)^{q+1}\right\}.$$

For every restriction, the target is parity or its negation on the m remaining coordinates. Lemma 6.1 gives error at least

$$B(m,q) := 2^{-m}\sum_{j=0}^{\lfloor (m-q-1)/2\rfloor}\binom{m}{j}$$

for every degree-q polynomial threshold function on that cube. This statement holds for real coefficients: its proof uses real linear algebra. Ties can be assigned either fixed sign by adding an arbitrarily small constant. Whenever z and Q have different signs, the global margin of z implies $|z - Q| \ge a$. Averaging the resulting error bound proves (21).

For completeness, choose a sufficiently small absolute $\delta > 0$, put $t = k/u$, and, when t exceeds a sufficiently large absolute constant, set $m = \min\{k, \lfloor \delta t^2 \rfloor\}$. If $m = k$, then $u \le \sqrt{\delta k}$; use $Q = P$ directly. The central binomial atom bound yields $B(k,u) \ge 7/16$ after increasing the fixed lower threshold on t. If $m < k$, take $q = \lfloor \sqrt{m}/32 \rfloor$. For example $\delta = (1024e)^{-2}$ ensures

$$\frac{eum}{k(q+1)} \le 32e\sqrt{\delta} \le \frac{1}{32}, \qquad q + 1 \ge c_0 t, \qquad B(m,q) \ge \frac{7}{16}$$

for sufficiently large t, with $c_0 > 0$ absolute. This proves (20) in that range. For bounded t, use $\|P\|_2^2 + \|T\|_2^2 = \|z\|_2^2 \ge a^2$ and enlarge C so that the asserted lower bound is nonpositive. □

In particular, any degree-u polynomial with global margin a whose threshold has parity error at most $3/8$ must satisfy

$$\|P\|_2^2 \geq c' a^2 \exp(ck/u).$$

For a fixed degree fraction u/k, this lower bound is a constant. It is therefore consistent with the bounded-amplitude blocked construction in Proposition 12.1.

8 Random tail covariance matrices

Let Q be the covariance kernel retaining only degrees greater than an odd cutoff r, and write $\tau = \tau_r$, $\ell = \lambda_{r+1}$, and $s_2 = s_{2,r}$. This section proves the sample event needed later, without assuming that a Gaussian approximation remains valid after conditioning. The entire argument concerns the exact limiting Gaussian covariance.

Lemma 8.1 (Far-pair decay)**.** *There are constants $c_0, C > 0$ such that, when $r \leq D/8$ is odd,*

$$\max_{d/4 \leq \mathrm{dist}(x,y) \leq 3d/4} |Q(x,y)| \leq \rho_r := C\sqrt{D}\, e^{-r/8} + Ce^{-c_0 D}.$$

Proof. The full-cube Krawtchouk polynomial satisfies

$$\sum_{j=0}^{D} \mathsf{K}_j(h; D) z^j = (1+z)^{D-h}(1-z)^h.$$

If $\delta D \leq h \leq (1-\delta)D$, put $b = \min(h, D-h)$. On $|z| = t$ the generating function is bounded in absolute value by

$$(1+t^2)^b (1+t)^{D-2b} \leq (1+t)^D \exp\left[-\frac{2\delta D t}{(1+t)^2}\right].$$

Cauchy's coefficient bound at $t = j/(D-j)$ and the uniform Stirling lower bound for $\binom{D}{j}$ therefore give, for $1 \leq j < D$,

$$\frac{|\mathsf{K}_j(h; D)|}{\binom{D}{j}} \leq C\sqrt{D} \exp[-2\delta j(1 - j/D)]. \tag{22}$$

For reduced-cube points at distance in $[d/4, 3d/4]$, their augmented full-cube distance lies in $[D/8, 7D/8]$ for large d. Because r is odd, the exact tail is $Q(x,y) = \sum_{j>r,\ j \text{ odd}} a_j \mathsf{K}_j(h; D)$. For $r < j \leq D - r$, (22) gives a bound $C\sqrt{D}e^{-r/8}$ after summation, since the total spectral mass is one. For $j > D - r$, monotonicity and an odd cutoff b_D near $D/4$ give

$$\sum_{j>D-r} \binom{D}{j} a_j \leq a_{b_D} \sum_{k<r} \binom{D}{k} \leq CD^{-\beta-1/2} \exp[-D(h(1/4) - h(1/8)) + O(1)].$$

Here $h(t) = -t \log t - (1-t)\log(1-t)$ denotes the binary entropy in nats. The positive entropy gap absorbs the polynomial factor and proves the claim. Adjust constants for bounded d. □

Lemma 8.2 (A tail Gram deviation bound)**.** *Let $X_1, \ldots, X_n$ be independent uniform cube points, with $1 \ll n \leq M/4$. Let*

$$B_{\mathrm{near}} = \#\{y : \mathrm{dist}(x,y) \notin [d/4, 3d/4]\} \leq 2e^{dh(1/4)}, \qquad q = 2\log M.$$

The count is independent of x. For every $z > 0$,

$$\mathbb{P}\{\|Q_{SS} - \tau I_n\|_{\mathrm{op}} > z\tau\} \\ \le \frac{C}{z^2}\left[\frac{q\rho_r}{\tau} + \frac{\sqrt{nqs_2}}{\tau} + \frac{n\ell}{\tau} + \frac{n(1+B_{\mathrm{near}})}{M}\right]^2 + \frac{n^2(1+B_{\mathrm{near}})}{M}. \tag{23}$$

Here the sampled matrix includes repeated training locations.

Proof. Let $J = Q - \tau I_M$, a zero-diagonal Hermitian matrix. Form a deterministic mask J_{far} retaining only pairs at distances in $[d/4, 3d/4]$. Since Q is positive semidefinite with constant diagonal τ, every entry has absolute value at most τ. Thus

$$\|J_{\mathrm{far}}\|_{\max} \le \rho_r, \\ \|J_{\mathrm{far}}\|_{1\to 2}^2 \le Ms_2, \\ \|J_{\mathrm{far}}\|_{\mathrm{op}} \le M\ell + \tau + \tau B_{\mathrm{near}}.$$

For the second inequality, Parseval gives squared column norm $Ms_2 - \tau^2$ for J, and entrywise masking only decreases it. For the third, $\|J\| \le M\ell + \tau$, while the removed matrix has maximum absolute row sum at most τB_{near}. The entropy bound on B_{near} is the elementary binomial tail bound on distances to a fixed cube point.

Let R retain each of the M coordinates independently with probability $p = 2n/M$. The zero-diagonal case of Tropp's random principal-submatrix theorem [4, Theorem 1.1], at moment order two and $q = 2\log M$, states

$$(\mathbb{E}\|RJR\|^2)^{1/2} \le C\left[q(\mathbb{E}\|RJR\|_{\max}^2)^{1/2} + \sqrt{pq}(\mathbb{E}\|JR\|_{1\to 2}^2)^{1/2} + p\|J\|\right].$$

Apply it to J_{far} and substitute the preceding bounds:

$$(\mathbb{E}\|RJ_{\mathrm{far}}R\|^2)^{1/2} \le C[q\rho_r + \sqrt{nqs_2} + n\ell + (n/M)\tau(1+B_{\mathrm{near}})]. \tag{24}$$

The logarithm here is that of the original dimension M, not the sample size.

To pass to a uniform subset of size n, use a uniform permutation of all cube points, independently of $B \sim \mathrm{Bin}(M, 2n/M)$. Its first B elements form the Bernoulli subset. On $B \ge n$, the first n principal submatrix is a compression of the larger one, so its operator norm cannot be larger. The size-n event is independent of B, and $\mathbb{P}(B \ge n) \ge 1/2$ for large n. Thus Markov's inequality applied to (24) gives the same fixed-size probability bound with a constant factor loss.

Conditional on no collisions, an independent size-n sample is a uniform size-n subset, up to an irrelevant ordering. Collisions have probability at most $n(n-1)/(2M)$. The probability of any sampled pair violating the distance condition is at most $n^2 B_{\mathrm{near}}/M$. On their complement, the sampled matrices J and J_{far} agree. Adding these exceptional probabilities proves (23). □

9 Tail covariance estimates at sparse cutoffs

The original principal-submatrix estimate uses the logarithm of the full cube cardinality. We retain that estimate here. A more precise bound on entries near orthogonal input pairs makes it applicable even when the spectral cutoff is a sufficiently large fixed integer.

Lemma 9.1 (Near-orthogonal entries)**.** *There are constants $a_0, \delta_0, c, C > 0$ such that the following holds. Put $D = d+1$, let r be odd with $r+2 \le \delta_0 D$, and let h be the Hamming distance between the augmented inputs $(1,x), (1,y)$. If $a = |D-2h|/D \le a_0$, then*

$$\frac{|Q_r(x,y)|}{\tau_r} \le [C(a + \sqrt{(r+2)/D})]^{r+2} + Cd^{\beta}e^{-cd}, \qquad Q_r = K_{>r}. \tag{25}$$

Proof. For $1 \le j < D$, the Krawtchouk generating function gives

$$\frac{|\mathsf{K}_j(h;D)|}{\binom{D}{j}} \le [e^{3/2}(a+\sqrt{j/D})]^j. \tag{26}$$

Indeed, set $b = \min(h, D-h)$. On the circle $|z| = t$, the generating function has modulus at most

$$(1+t^2)^b(1+t)^{D-2b} \le \exp(Dt^2/2 + Dat).$$

Cauchy's coefficient bound and $\binom{D}{j} \ge (D/j)^j$, with $t = j/[D(a+\sqrt{j/D})]$, prove (26): both Dt^2 and Dat are at most j.

Choose a_0, δ_0 so small that $e^{3/2}(a_0+\sqrt{\delta_0}) \le e^{-1}$ and $\delta_0 < 1/8$. On $j \le \delta_0 D$, the derivative of $j\log[e^{3/2}(a+\sqrt{j/D})]$ is at most $-1+1/2 < 0$. Consequently (26) is bounded by its value at $r+2$ throughout $r < j \le \delta_0 D$. The sum of the corresponding positive spectral masses $\binom{D}{j}a_j$ is at most τ_r. For $\delta_0 D < j \le (1-\delta_0)D$, the central-distance estimate (22) bounds the normalized Krawtchouk polynomial by $C\sqrt{D}e^{-cD}$; its polynomial factor can be absorbed by decreasing c. For $j > (1-\delta_0)D$, choose an odd b_D near $D/4$. Monotonicity of a_j and Lemma 3.2 bound the remaining mass by

$$a_{b_D} \sum_{i<\delta_0 D} \binom{D}{i} \le \mathrm{poly}(D) e^{-D[\mathsf{h}(1/4)-\mathsf{h}(\delta_0)]},$$

where $\mathsf{h}(t) = -t\log t - (1-t)\log(1-t)$. The entropy gap is positive. Finally $\tau_r \ge cD^{-\beta}$ by Lemma 3.2, proving (25). □

Lemma 9.2 (Sparse adaptive tail Gram matrices)**.** *Fix $A \ge 1$, a failure probability $\delta > 0$, and an odd integer $r_0 \ge 11$. There is $\alpha_0 > 0$ such that, for any fixed $0 < \alpha \le \alpha_0$, the following assertions hold uniformly as $d \to \infty$. Suppose r is odd,*

$$r_0 \le r \le \alpha d, \qquad AD_r \log(D_r+1) \le n < AD_{r+2}\log(D_{r+2}+1).$$

Put $u = r+2$, $\rho = r/(d+1)$, and $H = \log(e/\rho)$. For each such specified n, with probability at least $1-\delta$,

$$\begin{aligned} \|(Q_r)_{SS}\|_{\mathrm{op}} &\le C_{L,\delta}\left[1 + \frac{\sqrt{nd\,s_{2,r}}}{\tau_r} + \frac{n\ell_r}{\tau_r}\right]\tau_r \\ &\le C_{L,\delta}[1+\sqrt{A\rho^{-1}H} + AH]\tau_r, \qquad (27) \\ \|(Q_u)_{SS} - \tau_u I_n\|_{\mathrm{op}} &\le [C_{L,\delta}(\sqrt{A\rho H} + A\rho^2 H) + o(1)]\tau_r. \qquad (28) \end{aligned}$$

The $o(1)$ is uniform over the stated range. In particular, after reducing α_0 if necessary, $\|(Q_u)_{SS}\|_{\mathrm{op}} \le 2\tau_r$ on the same event for all sufficiently large d. Deterministically,

$$\begin{aligned} 0 \le \tau_r - \tau_u &\le C_L\tau_r/r, \qquad (29) \\ n\ell_u/\tau_r &\le C_L A\rho^2 H, \qquad (30) \end{aligned}$$

where $\ell_v = \lambda_{v+1}$. No simultaneous event over different sample sizes is asserted.

Proof. The dimension and spectral estimates already proved give

$$\begin{aligned} n &\le CA\binom{D}{r+2}rH, & \log(2n) &\le C_A rH, \\ n\ell_r/\tau_r &\le C_L AH, & nd\,s_{2,r}/\tau_r^2 &\le C_L A\rho^{-1}H. \end{aligned}$$

For $u = r + 2$, the exact two-step binomial ratio gives

$$n\ell_u/\tau_r \le C_L A\rho^2 H, \qquad nd\, s_{2,u}/\tau_r^2 \le C_L A\rho H.$$

Moreover $\tau_r - \tau_u = \binom{D}{r+2} a_{r+2}$, so the level and trace estimates prove (29).

We apply the proof of Lemma 8.2 with a different deterministic mask. Retain pairs satisfying

$$\left|\frac{1 + x \cdot y}{d+1}\right| \le a_n, \qquad a_n = \sqrt{128 \log(2n)/d}.$$

By choosing α_0 sufficiently small, uniformly $\log(2n)/d \le C_A \alpha \log(e/\alpha) + o(1)$ is as small as needed, so $a_n \le a_0$. For independent uniform inputs, Hoeffding's bound (and $a_n \ge 2/(d+1)$) gives exceptional-pair probability

$$b_n \le 2\exp(-a_n^2 d/8) = 2(2n)^{-16}.$$

The exceptional set is translation invariant on the cube. Its size is therefore the same in every row, so the masked-matrix proof applies with $B_{\mathrm{near}}/M = b_n$. This mask is deterministic for each fixed pair (d, n); it does not depend on the observed sample. Lemma 9.1 supplies the retained-entry bound

$$\frac{\eta_v}{\tau_v} \le \left[C\sqrt{\frac{\log(2n) + v + 2}{d}}\right]^{v+2} + Cd^{\beta} e^{-cd}, \qquad v \in \{r, u\}. \tag{31}$$

Thus exactly the argument of Lemma 8.2, still with $q = 2\log M = 2d\log 2$, gives for every $z > 0$

$$\begin{aligned}&\mathbb{P}\{\|(Q_v)_{SS} - \tau_v I_n\| > z\tau_v\} \\ &\quad \le \frac{C}{z^2}\left[\frac{q\eta_v}{\tau_v} + \frac{\sqrt{nqs_{2,v}}}{\tau_v} + \frac{n\ell_v}{\tau_v} + n(M^{-1} + b_n)\right]^2 + n^2(M^{-1} + b_n).\end{aligned} \tag{32}$$

The collision and exceptional-pair terms tend to zero uniformly: $n \ge AD_{r_0}\log(D_{r_0}+1) \to \infty$, and α_0 can be chosen so that $n \le e^{c_1 d}$ with $2c_1 < \log 2$.

The coherence term $q\eta_v/\tau_v$ also tends to zero uniformly. For $r \le \sqrt{d}$, its bracket in (31) is at most $C_A d^{-1/4}\sqrt{\log d}$, and its exponent is at least $r_0 + 2 \ge 13$. Hence it is bounded, after multiplication by q, by

$$Cd[C_A d^{-1/4}\sqrt{\log d}]^{13} + Cd^{1+\beta}e^{-cd} = o(1).$$

For $r > \sqrt{d}$, reducing α_0 makes the bracket at most $1/2$, so the bound is $Cd\,2^{-\sqrt{d}} + Cd^{1+\beta}e^{-cd} = o(1)$. Using (32) for $v = r, u$, with failure probability at most $\delta/2$ for each, now proves (27) and (28). For the latter, multiply by $\tau_u/\tau_r \le 1$ before using the spectral bounds, so no lower comparison of the two traces is needed. Finally $\rho H \to 0$ as $\rho \to 0$, uniformly for $0 < \rho \le \alpha$; choose α_0 to make the fixed part of the fine deviation below $1/2$, and absorb the uniform $o(1)$. Since $\tau_u \le \tau_r$, this gives the asserted upper bound $2\tau_r$. □

10 An exponential sample lower bound

Throughout this section $f = \chi_U$, where $U \subseteq [d]$ and $|U| = k$. Write $D = d + 1$, $\beta = 2^{-(L-1)}$, and

$$B_{d,k} = \sqrt{k}\log\frac{eD}{\sqrt{k}}, \qquad W_{d,k} = \max\{k, B_{d,k}\}.$$

The k term follows from the random-restriction amplitude tradeoff. The second term is useful in very large ambient dimensions and follows from a direct polynomial-threshold obstruction.

Theorem 10.1 (Sparse parity for $0 < \gamma < 1$)**.** *Fix $L \geq 2$ and $a > 0$. There are constants $c > 0$, k_0, and d_0, depending only on L, a, such that the following holds. If $d \geq d_0$, $k_0 \leq k \leq d$, and a specified sample size satisfies*

$$0 \leq n \leq \exp(cW_{d,k}),$$

then, with probability at least $3/4$ over independent uniform training inputs, every minimizer of

$$\min\left\{\frac{1}{2}\|z\|_K^2 : |z(x)| \geq a \text{ for all } x, \quad f(X_i)z(X_i) \geq a \text{ for all } i\right\}$$

has population sign error at least $3/8$. On the same event, for every fixed $0 < \gamma < 1$, the actual network posterior in the width-first limit satisfies

$$\liminf_{N\to\infty} \mathbb{P}_x\{q_{N,S}(x) \leq 2/3\} \geq \frac{1}{16}.$$

The probability statement is for each specified sample size; it is not a simultaneous event over all sample prefixes.

In particular, the lower bound is exponential in k throughout the entire growing sparse regime. It does not give the stronger scale $\exp\{ck\log(ed/k)\}$ when $d/k \to \infty$.

10.1 Squared balance on the target coordinates

Lemma 10.2 (Active-coordinate squared balance)**.** *If P has total Walsh degree at most r, then*

$$\mathbb{E}(fP)_-^2 \geq \left(\frac{1}{2} - \sqrt{\frac{r}{k}}\right)\|P\|_2^2. \tag{33}$$

For every fixed $\delta > 0$, there is a universal A_δ such that if $r \leq k/64$ and $n \geq A_\delta D_r \log(D_r + 1)$, then, with probability at least $1 - \delta$,

$$\mathbb{P}_n(fP)_-^2 \geq \frac{1}{4}\|P\|_2^2 \quad \text{for every polynomial } P \text{ of degree at most } r. \tag{34}$$

Proof. For $j \in U$ put $D_jP = (P - P \circ \mathrm{flip}_j)/2$. Because f changes sign under these flips, the inequality $|F(s) - F(t)| \leq (|s| + |t|)|s - t|$ for $F(t) = t|t|$ gives

$$|\mathbb{E}fP|P|| \leq 2\|P\|_2\|D_jP\|_2.$$

Averaging over $j \in U$ and using $\sum_{j\in U}\|D_jP\|_2^2 \leq r\|P\|_2^2$ proves (33) from $(fP)_-^2 = (P^2 - fP|P|)/2$. The vector of all Walsh characters of degree at most r is isotropic and has squared length D_r at every input. Under $r \leq k/64$, the population bound is at least $3\|P\|_2^2/8$. Apply Lemma 5.3 at accuracy $1/8$ and use homogeneity to obtain (34). □

10.2 The polynomial-threshold range

Lemma 10.3 (A $\sqrt{k}$ cutoff)**.** *For fixed L, a, there are $c_b > 0$, k_b, and d_b such that, on the localized-comparator event, every minimizer has error at least $3/8$ whenever $d \geq d_b$, $k_b \leq k \leq d$, and*

$$n \leq \exp\left(c_b\sqrt{k}\log\frac{eD}{\sqrt{k}}\right).$$

Proof. The comparator gives the simultaneous bound

$$\|z\|_K^2 \le C_{L,a}(d + nd^\beta) \tag{35}$$

for every minimizer. Choose a sufficiently small constant $\kappa > 0$ and an odd s with $s \sim \kappa\sqrt{k}$. Fixing the $d-k$ nuisance coordinates leaves a degree-s polynomial on the active k-cube. Lemma 6.1 and the central binomial bound imply that every such polynomial threshold has error at least $7/16$, for all sufficiently large k.

The sharp spectrum gives

$$\|z - z_{\le s}\|_2^2 \le C_{L,a}(d + nd^\beta)\lambda_{s+1} \le \frac{C_{L,a}(d + nd^\beta)s^{-1-\beta}}{\binom{D}{s+2}}.$$

The binomial lower bound shows

$$\log \binom{D}{s+2} \ge c_1\kappa\sqrt{k}\log\frac{eD}{\sqrt{k}}$$

for a universal $c_1 > 0$ once k is large. Choose $c_b < c_1\kappa/4$, then increase k_b to absorb the factor d^β and increase d_b to make the displayed tail bound at most $a^2/16$, uniformly in the stated range. The additive d is harmless because s is then a sufficiently large fixed integer or larger. Since $|z| \ge a$, its sign differs from a fixed-tie threshold of $z_{\le s}$ on at most $1/16$ of the cube. Its error is therefore at least $7/16 - 1/16 = 3/8$. The comparator's allowed sample range contains the present one for all large d, because $\sqrt{k}\log(eD/\sqrt{k}) = o(d)$ uniformly in $k \le d$. □

10.3 Proof of the positive-subcritical theorem

Proof of Theorem 10.1. Fix $\delta = 1/40$, choose A large enough for Lemma 10.2, and fix an odd $r_0 \ge 11$. Choose the small degree fraction in Lemma 9.2 after these choices. The localized comparator has failure probability $e^{-\Omega_L(d)}$. All its conclusions hold simultaneously over the minimizing fields.

If $W_{d,k} = B_{d,k}$, choose $c \le c_b$ and apply Lemma 10.3. We may therefore assume $W_{d,k} = k$ and $n \le e^{ck}$. Let r be the largest odd integer satisfying

$$AD_r\log(D_r + 1) \le n.$$

If this cutoff does not exist or $r < r_0$, then $n < AD_{r_0}\log(D_{r_0} + 1)$. After increasing k_0, this entire range lies below the threshold of Lemma 10.3, which again proves the claim.

Suppose $r \ge r_0$. Choose $c < (\log 2)/4$. Since $k \le d$ and $D_r \ge 2^{d-1}$ for $r \ge d/2$, the defining lower bound on n excludes $r \ge d/2$ for sufficiently large d. In this range, the binomial lower bound implies

$$rH \le C_0\log n \le C_0 ck, \qquad \rho = \frac{r}{D}, \quad H = \log\frac{e}{\rho}, \tag{36}$$

for a universal constant C_0. By taking c sufficiently small, we ensure $r \le k/64$, $r \le \alpha d$, and $n \le e^{d/25}$. All comparator and sparse-Gram hypotheses therefore hold. Moreover $r \ge r_0$ makes n polynomially larger than d, so (35) becomes

$$B := \|z\|_K^2 \le C_{L,a}nd^\beta. \tag{37}$$

Work on the intersection of the comparator event, the squared-sampling event (34), and the Gram event in Lemma 9.2. Their total failure probability is at most $2\delta + e^{-\Omega_L(d)} < 1/4$ for sufficiently large d. For an arbitrary minimizer write

$$z = P + b + t, \qquad P = z_{\le r}, \quad b = z_{(r,r+2]}, \quad t = z_{>r+2}.$$

At a training point where $fP < 0$, consistency gives $(fP)_- \le |z - P|$. The sampling bound, RKHS evaluation bound, and (27) yield

$$\frac{1}{4}\|P\|_2^2 \le \mathbb{P}_n(fP)_-^2 \le \mathbb{P}_n(z-P)^2 \le \frac{B}{n}\|(Q_r)_{SS}\|_{\mathrm{op}}.$$

Using (37), $\tau_r \asymp r^{-\beta}$, and the deterministic estimates $n\lambda_{r+1} \le C_L AH\tau_r$ and $n\lambda_{r+3} \le C_L A\rho^2 H\tau_r$, we obtain, simultaneously for every minimizer,

$$\|P\|_2^2 \le C_{L,a,A,\delta}\rho^{-(\beta+1)}H, \tag{38}$$

$$\|b\|_2^2 \le B\lambda_{r+1} \le C_{L,a,A}\rho^{-\beta}H, \tag{39}$$

$$\|t\|_2^2 \le B\lambda_{r+3} \le C_{L,a,A}\rho^{2-\beta}H. \tag{40}$$

For the first estimate we bounded $1 + \sqrt{A\rho^{-1}H} + AH$ by $C_A\rho^{-1}H$. The eigenvalue estimates follow from the sharp spectrum and the maximality interval for r; the second is also stated in (30). Put $h = P + b = z_{\le u}$, where $u = r + 2 \le 2r$. The first two pieces are orthogonal in L^2, and their sum therefore has squared norm at most $C\rho^{-(\beta+1)}H$. Hence Lemma 7.1 gives

$$\mathbb{P}\{\operatorname{sign} z \ne f\} \ge \frac{7}{16} - C_1\rho^{2-\beta}H - C_1\rho^{-(\beta+1)}H\exp\left(-c_*\frac{k}{u}\right), \tag{41}$$

where $c_* > 0$ is universal and C_1 depends only on the already fixed constants L, a, A, δ.

We now choose the theorem's c. From (36), $\rho \le C_0 c$ and $k/u \ge H/(2C_0 c)$. Since $x^{2-\beta}\log(e/x) \to 0$ as $x \downarrow 0$, the first error term in (41) is at most $1/32$ once c is sufficiently small. For the second term, use $\rho^{-(\beta+1)} = \exp[(\beta+1)(H-1)]$ to obtain

$$C_1\rho^{-(\beta+1)}He^{-c_*k/u} \le C_1 H\exp\left[\left(\beta + 1 - \frac{c_*}{2C_0 c}\right)H\right].$$

As $H \ge 1$, taking c still smaller makes this at most $1/32$ uniformly. These choices also ensure all earlier degree-fraction restrictions. Finally increase k_0, d_0 to absorb finite rounding and uniform asymptotic estimates. Thus every minimizing sign has error at least $7/16 - 1/32 - 1/32 = 3/8$.

For completeness, let $q(x)$ be any probability mixture of these minimizing signs, let $R = \mathbb{E}_x(1 - q(x))$, and put $F = \mathbb{P}_x\{q(x) \le 2/3\}$. The draw-error bound gives $R \ge 3/8$, while

$$R \le \frac{1-F}{3} + F, \qquad F \ge \frac{3R-1}{2} \ge \frac{1}{16}.$$

The positive-gamma width-limit proposition says that, for each fixed $0 < \gamma < 1$, the actual finite-width posterior puts mass $1 - o(1)$ on these minimum-energy signs. Thus its draw error is at least $3/8 - o(1)$, independently of the relative weights among ties. Apply the same inequality at every width and take the lower limit. This proves the asserted actual-posterior guarantee. □

11 An upper bound and the dependence on ambient dimension

An upper bound identifies the natural comparison scale in the sparse regime.

Proposition 11.1 (An actual-posterior upper bound)**.** *Let λ_k be the normalized Walsh eigenvalue defined above. There is an absolute constant C such that, for every $0 < \varepsilon < 1/3$, $0 < \delta < 1$, and*

$$n \ge \frac{C}{\varepsilon^2}\left(\lambda_k^{-1} + \log\frac{2}{\delta}\right),$$

the actual posterior at every fixed $0 < \gamma < 1$ *satisfies*

$$\mathbb{P}_S\left\{\limsup_{N\to\infty} F_N(S) \le 3\varepsilon\right\} \ge 1-\delta.$$

On the same sample event, every compatible minimum-energy sign function has test error at most ε.

Proof. The globally separated comparator $z = af$ is feasible for every sample and has $\|af\|_K^2 = a^2/\lambda_k$. Thus every minimizing field lies in the RKHS ball of radius $R = a/\sqrt{\lambda_k}$ and satisfies $f(X_i)z(X_i) \ge a$. Use the ramp loss $\ell(t) = \min\{1, \max\{0, 1 - t/a\}\}$. Its empirical loss vanishes for these fields, and $\mathbf{1}_{\{f(x)z(x)\le 0\}} \le \ell(f(x)z(x))$.

For completeness, the standard uniform deviation estimate here is

$$\sup_{\|z\|_K\le R}\left[\mathbb{E}\ell(fz) - \frac{1}{n}\sum_i \ell(f(X_i)z(X_i))\right] \le \frac{4R}{a\sqrt{n}} + \sqrt{\frac{\log(1/\delta)}{2n}}$$

with probability at least $1-\delta$. Indeed, symmetrization and contraction for the a^{-1}-Lipschitz function $\ell - \ell(0)$ bound the expected supremum by $4/a$ times the RKHS Rademacher complexity. The reproducing identity and $K(x,x) = 1$ give

$$\frac{1}{n}\mathbb{E}_\sigma \sup_{\|z\|_K\le R}\left|\sum_i \sigma_i f(X_i)z(X_i)\right| = \frac{R}{n}\mathbb{E}_\sigma\left\|\sum_i \sigma_i f(X_i)K_{X_i}\right\|_K \le \frac{R}{\sqrt{n}}.$$

Changing one sample changes the supremum deviation by at most $1/n$; the bounded-differences inequality supplies the displayed probability bound. The assumed sample size therefore makes every minimizing sign's error at most ε.

By Proposition 2.1, the actual posterior mass outside these minimizing signs tends to zero. Hence its average test error $\mathbb{E}_x[1 - q_{N,S}(x)]$ has upper limit at most ε. Since $F_N(S) \le 3\mathbb{E}_x[1 - q_{N,S}(x)]$, the claimed confidence bound follows, including the strict greater-than-2/3 convention. □

For growing $k = o(d)$, put $j = k$ if k is odd and $j = k+1$ otherwise. Lemma 3.2 gives

$$\lambda_k^{-1} \asymp j^{1+\beta}\binom{d+1}{j}, \qquad \log\lambda_k^{-1} = \Theta\left(k\log\frac{ed}{k}\right).$$

In particular, when $k = \lfloor d/\log d\rfloor$, fixed $\varepsilon, \delta > 0$ require only $\exp\{O(k\log\log d)\} = d^{o(k)}$ samples in this upper bound. Thus a literal $d^{\Omega(k)}$ lower bound, with a fixed positive exponent uniformly over all growing sparse $k = o(d)$, is false for the actual posterior. The upper bound has the natural exponential scale $\exp\{\Theta(k\log(ed/k))\}$. A lower bound on this scale remains open. For $k \le d^{1-\eta}$ with fixed $\eta > 0$, such a lower bound would entail $d^{\Omega(k)}$.

12 Why an amplitude bound alone cannot settle the sparse case

The remaining difficulty cannot be removed simply by proving that every accurate low-degree threshold approximation to parity has amplitude growing with the number of active coordinates. The following construction rules out that assertion without any symmetry assumption on the polynomial.

Proposition 12.1 (Bounded-amplitude blocked approximants)**.** *For every $\alpha, \varepsilon > 0$, there are finite constants $B = B(\alpha, \varepsilon)$ and $k_0 = k_0(\alpha, \varepsilon)$ such that, for every $d \ge k \ge k_0$, a multilinear polynomial P on the d-cube satisfies*

$$\deg P \le \alpha k, \qquad 1 \le |P(x)| \le B \quad \text{for every } x, \qquad \Pr\{\operatorname{sign} P(X) \ne \chi_U(X)\} \le \varepsilon$$

for any fixed $U \subseteq [d]$ of size k.

Proof. Replacing α, ε by smaller positive constants if necessary, assume $\alpha \le 1$ and $\varepsilon \le 1/2$. Choose a fixed integer m sufficiently large that a central integer interval $[A, B_0]$ for $T \sim \mathrm{Bin}(m, 1/2)$ has probability at least $1 - \varepsilon$ and length $s = B_0 - A \le \alpha m/2$. Hoeffding's inequality permits $m = O(\alpha^{-2} \log(2/\varepsilon))$, after increasing the absolute constant. Define

$$Q(t) = \sigma \prod_{j=A}^{B_0 - 1} (2t - 2j - 1),$$

where $\sigma \in \{-1, 1\}$ makes its sign agree with $(-1)^t$ throughout $[A, B_0] \cap \mathbb{Z}$. At every integer $t \in [0, m]$, all factors are nonzero odd integers, so $1 \le |Q(t)| \le (2m+1)^s$. Partition U into m nonempty blocks of size at most $\lceil k/m \rceil$ and let Y_j be the parity of the jth block. These are independent uniform signs, and their product is χ_U. Set

$$P(x) = Q\left(\frac{m - \sum_{j=1}^{m} Y_j(x)}{2} \right).$$

Multilinearization in the block variables shows $\deg P \le s\lceil k/m \rceil \le \alpha k$ whenever $k \ge m$. The remaining conclusions follow from the bounds on Q and the binomial law of its argument. □

For example, if 3 divides k, three equal blocks give $P = -(Y_1 + Y_2 + Y_3)$, with degree $k/3$, global margin one, parity error $1/4$, and $\|P\|_2^2 = 3$. If 4 divides k, the polynomial $P = 1 - \sum_{j=1}^{4} Y_j$ has degree $k/4$, error $5/16$, and $\|P\|_2^2 = 5$. These examples need not be symmetric in all k original coordinates.

Proposition 12.1 is an obstruction to a proof method, not a learning upper bound for the posterior. If $\deg P < k$, then $\mathbb{E}[\chi_U P] = 0$; together with a global unit margin and sign error R, this implies

$$\mathbb{E}(1 - \chi_U P)_+^2 \ge \frac{1}{R}.$$

Thus an accurate blocked approximant has larger squared repair loss than the constant polynomial, whose loss is exactly two. Determining the minimum energy of repairs that also preserve the global margin remains essential.

13 A small bounded-weight parity network

For completeness, full parity on k bits has a one-hidden-layer tanh realization of width $k+1$, with weights $O(k \log(k+1))$ and output margin greater than $1/2$. Let

$$t(x) = \frac{k + \sum_i x_i}{2} \in \{0, \dots, k\}, \quad y_j = (-1)^{k-j}, \quad A = \log(16(k+1)).$$

The step functions $s_j(t) = \operatorname{sign}(t - j + 1/2)$ satisfy

$$y_t = c + \sum_{j=1}^{k} y_j s_j(t), \qquad c = (y_0 + y_k)/2 \in \{0, 1\}.$$

Replace each step by $h_j = \tanh(A(t - j + 1/2))$. Since its argument has absolute value at least $A/2$, its error is at most $2e^{-A}$. A constant neuron $h_0 = \tanh 1$ supplies the term c using output weight $c/\tanh 1$. The resulting output differs from parity by less than $1/8$. First-layer input weights are $A/2$ and bias weights have magnitude at most $Ak/2$; readout weights have magnitude at most 2.

At depth $L \geq 3$, the next hidden layer sums these features and applies tanh, producing the correct sign with magnitude at least $\tanh(7/8) > 1/2$. Each additional hidden layer can preserve this margin by a single weight 2, since $\tanh(2u) > 1/2$ whenever $u > 1/2$. A final readout weight 1 completes the construction. Unused neurons are set to zero. Thus the same polynomial bounds hold at every fixed depth. For any other fixed margin $a > 0$, multiplying the final readout by $2a$ gives margin greater than a with the same polynomial complexity bounds.

14 Scope

Theorem 10.1, together with the actual-network rate in Proposition 2.1, proves Theorem 1.1. Parity has a Boolean circuit with $O(k)$ gates and the bounded-weight neural realization given above. Its difficulty here is a property of the specified posterior. An algorithm that searches parity supports can identify the target with $O(k \log(ed/k))$ noiseless examples at fixed success probability: each incorrect parity agrees with the target on each independent uniform example with probability $1/2$, and a union bound applies to the candidate supports.

Section 11 gives the corresponding actual-posterior upper bound and explains the dependence on ambient dimension. A matching $\exp\{ck \log(ed/k)\}$ lower bound remains open in the growing sparse regime, including a $d^{\Omega(k)}$ lower bound when $k \leq d^{1-\eta}$ for fixed $\eta > 0$. No assertion about the endpoint $\gamma = 0$ is made in this paper.

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